

System Elements

1

Human decision makers access the ReactJS web application (Coverage, Scenario, Decision, and Executive Report views)

2

AWS WAF filters malicious traffic; CloudFront serves cached static assets from S3

3

Amazon Cognito authenticates users at Amazon API Gateway

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API Gateway routes authenticated requests into the orchestration layer

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Amazon EventBridge publishes application events to trigger workflows

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AWS Step Functions orchestrates multi-step analysis workflows

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AWS Lambda executes business logic, calling Bedrock and data stores

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AWS Lambda reads/writes AWS S3 (documents) and DynamoDB (records), with ElastiCache as a hot cache

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New S3 documents trigger Step Functions for re-analysis

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AWS Bedrock provides foundation models, embeddings, RAG retrieval, and Guardrails for safe AI inference

11

Embeddings are indexed in Amazon OpenSearch; RAG augments Bedrock prompts with retrieved context

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Bedrock generates Story, Coverage, Audience, Market, and Comparable Intelligence

13

Lambda computes Production Complexity and Financial Intelligence

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All seven intelligence signals feed the Explainable Verdict Engine

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Human reviewers review, annotate, and can override the AI verdict to reach a Final Decision

16

The Decision feeds the Scenario Simulation Engine, which re-runs the Verdict Engine under new assumptions

17

External sources (Market, Title/Content, Production, Talent, Financial data) enrich Market and Comparable Intelligence

18

CloudWatch, CloudTrail, IAM, Secrets Manager, and GuardDuty provide logging, auditing, and security across every layer

Service Color Key

Compute / API / Orchestration (Lambda, Step Functions, EventBridge, API GW)

Storage / Database (S3, DynamoDB, ElastiCache)

Networking / CDN & Decision Intelligence Layer

External Intelligence Sources

AI / ML Services (AWS Bedrock, OpenSearch)

Experience & Human-in-the-Loop Layer

Security & Observability

Architecture - Verdict AI - Explainable Greenlight Decisioning

The diagram illustrates the AWS architecture for Verdict AI, an explainable greenlight decisioning system. It is organized into several layers:

- Experience Layer - ReactJS Web Application:** Human Decision Makers interact with the ReactJS App (SPA), which provides views for Coverage, Scenario, Decision, and Executive Report.
- Application & API Layer:** AWS WAF filters malicious traffic. CloudFront serves cached static assets from S3. API Gateway routes authenticated requests (via Cognito) into the orchestration layer.
- Event-Driven Orchestration Layer:** Amazon EventBridge publishes application events to trigger workflows orchestrated by AWS Step Functions. AWS Lambda functions execute business logic, calling Bedrock and data stores.
- Data & Knowledge Layer:** AWS S3 (documents), DynamoDB (records), ElastiCache (hot cache), and OpenSearch (knowledge base) store and retrieve data.
- AI Intelligence Layer:** AWS Bedrock provides foundation models, embeddings, RAG retrieval, and Guardrails for safe AI inference. Amazon OpenSearch (vector index) is used for RAG.
- Security & Observability:** CloudWatch, CloudTrail, IAM, Secrets Manager, and GuardDuty provide logging, auditing, and security across every layer.
- Decision Intelligence Layer:** This layer contains the Explainable Verdict Engine and the Scenario Simulation Engine. It receives intelligence from various sources and generates a final decision.
- Human-in-the-Loop Layer:** Human Review, Annotations & Corrections, and Human Override are used to refine the AI-generated decision.
- External Intelligence:** Market Data, Title & Content Metadata, Production Data, Talent Data, and Financial Data are used to enrich the intelligence generated by the AI.

The flow of data and control is as follows:

- Human Decision Makers access the ReactJS App (SPA).
- The ReactJS App sends API Requests to API Gateway.
- API Gateway routes requests to AWS Lambda functions.
- AWS Lambda functions interact with AWS S3, DynamoDB, ElastiCache, and AWS Bedrock.
- AWS Lambda functions publish events to Amazon EventBridge.
- Amazon EventBridge triggers workflows orchestrated by AWS Step Functions.
- AWS Step Functions orchestrate the execution of AWS Lambda functions.
- AWS Lambda functions generate intelligence (Story, Coverage, Audience, Market, Comparable Intelligence, Production Complexity, Financial Intelligence).
- The intelligence is fed into the Explainable Verdict Engine and the Scenario Simulation Engine.
- The Explainable Verdict Engine generates a Final Decision.
- The Final Decision is reviewed by Human Review, which can lead to Annotations & Corrections or Human Override.
- The Human Review process can lead to a Final Decision or a Human Override.
- The Final Decision is used to generate an Executive Report.
- The Scenario Simulation Engine can re-run the Verdict Engine under new assumptions.
- External Intelligence (Market Data, Title & Content Metadata, Production Data, Talent Data, Financial Data) is used to enrich the intelligence generated by the AI.